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CHALLENGES FOR PHYSICS-BASED MACHINE LEARNING ARISING FROM BENCHMARK

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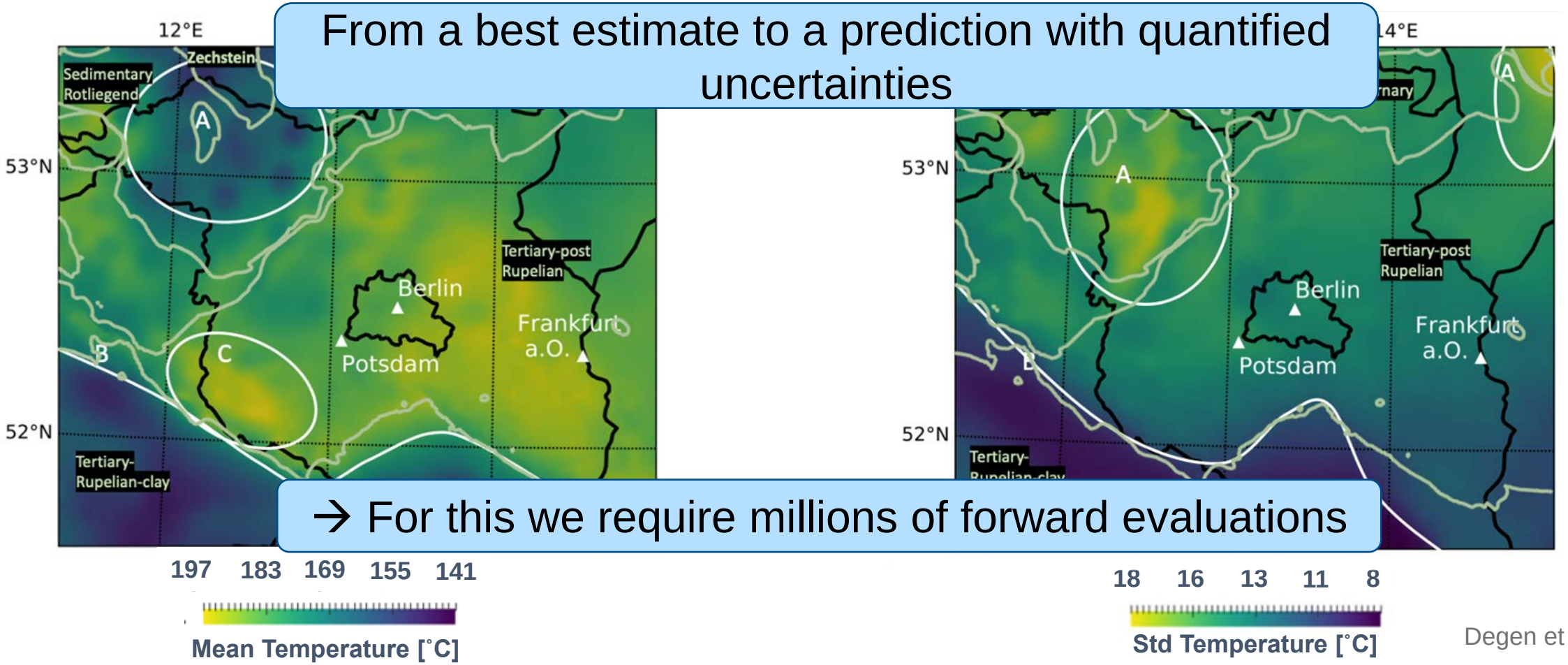
MOTIVATION

From a best estimate to a prediction with quantified uncertainties



→ For this we require millions of forward evaluations

Degen et al. (2022a)

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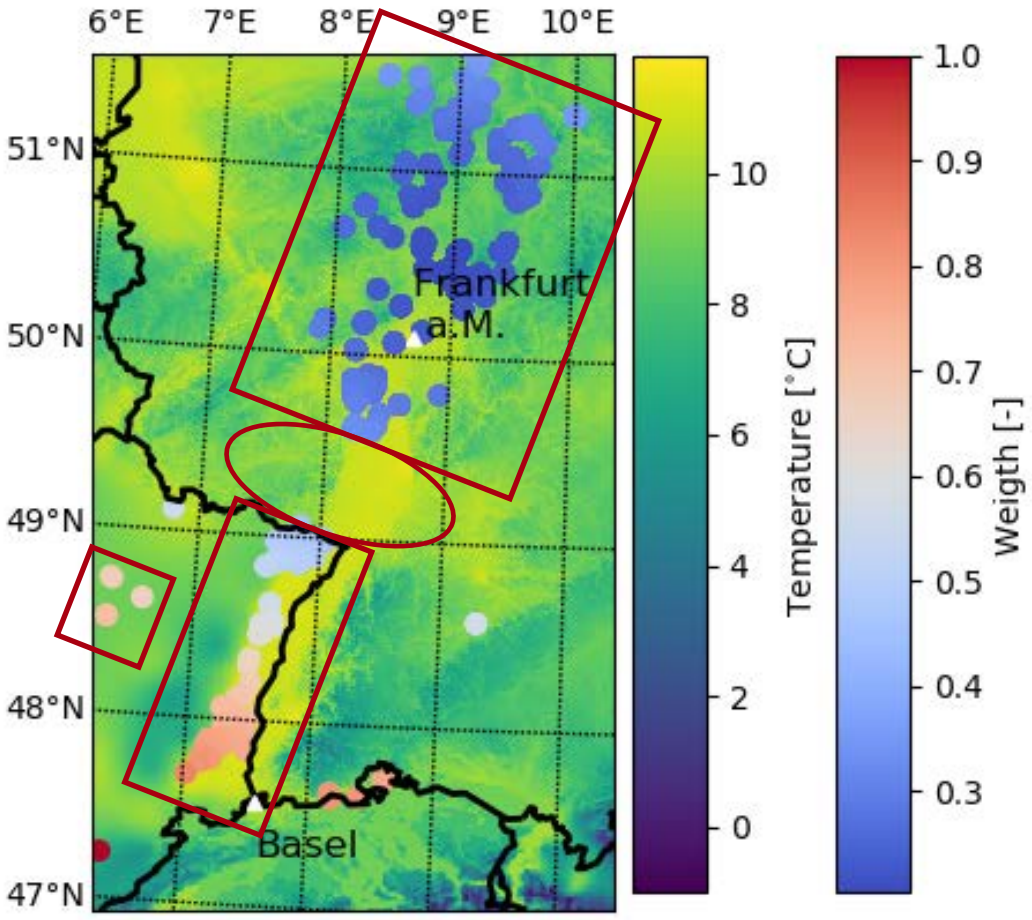
CHALLENGES

- Geophysical applications are characterized by:
  - Large scale spatial and temporal variability
  - Highly heterogeneous media
  - Complex coupled processes

→ Often nonlinear hyperbolic partial differential equations

- Data Sparsity because of
  - Cost
  - Technical Challenges
  - Confidentiality / Economical Interest
- Data Usability/Quality

→ High Uncertainties



Degen et al. (2021)

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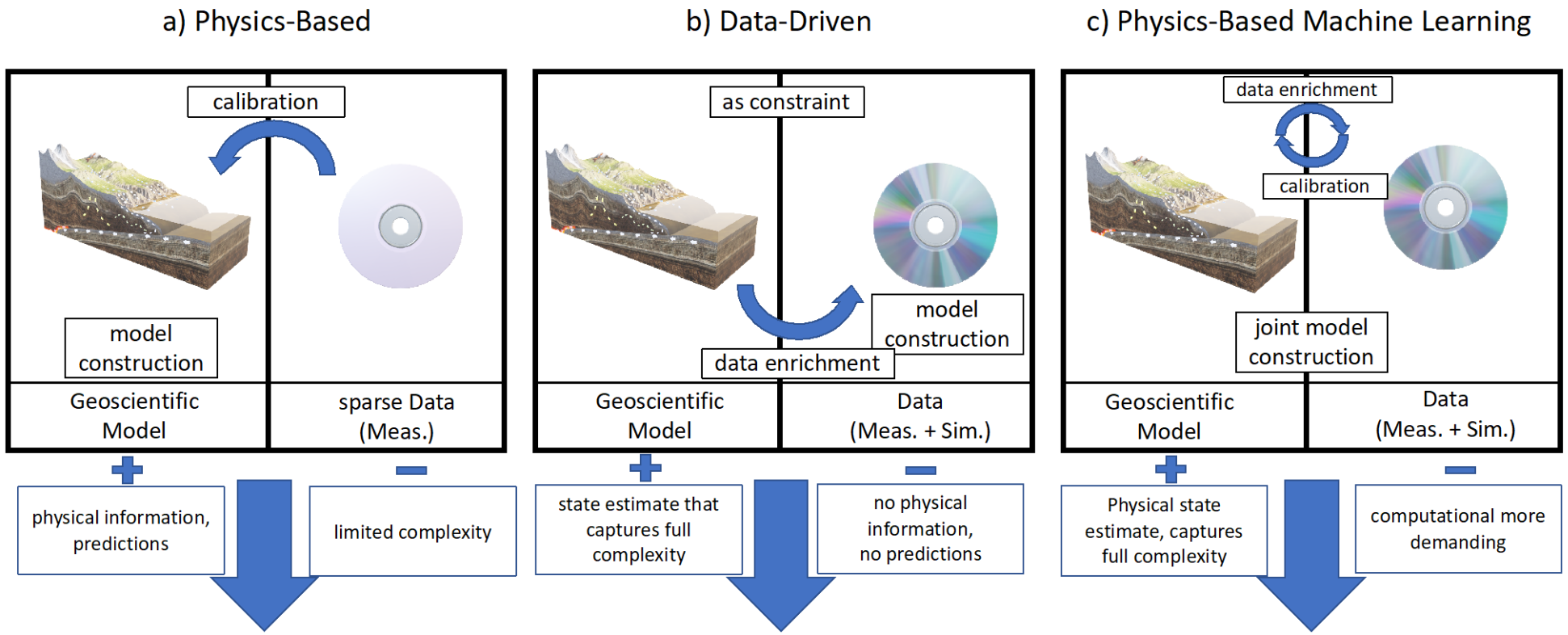
SURROGATE MODEL TECHNIQUES



Degen et al. (2023)

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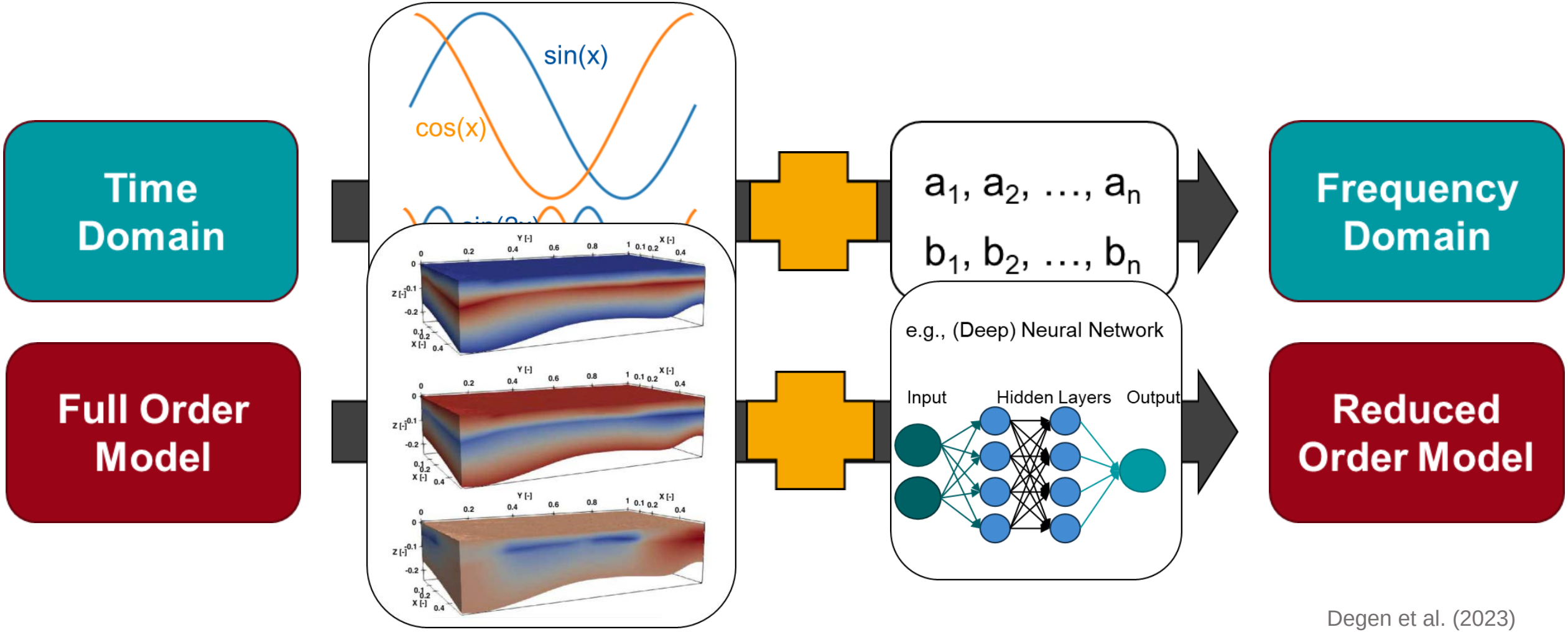
NON-INTRUSIVE REDUCED BASIS METHOD



Degen et al. (2023)

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NON-INTRUSIVE REDUCED BASIS METHOD

1. Construct a training and test data set
   - Latin Hypercube sampling for training set
   - Random sampling for test set

2. Scale the data-sets

   → training and test snapshots

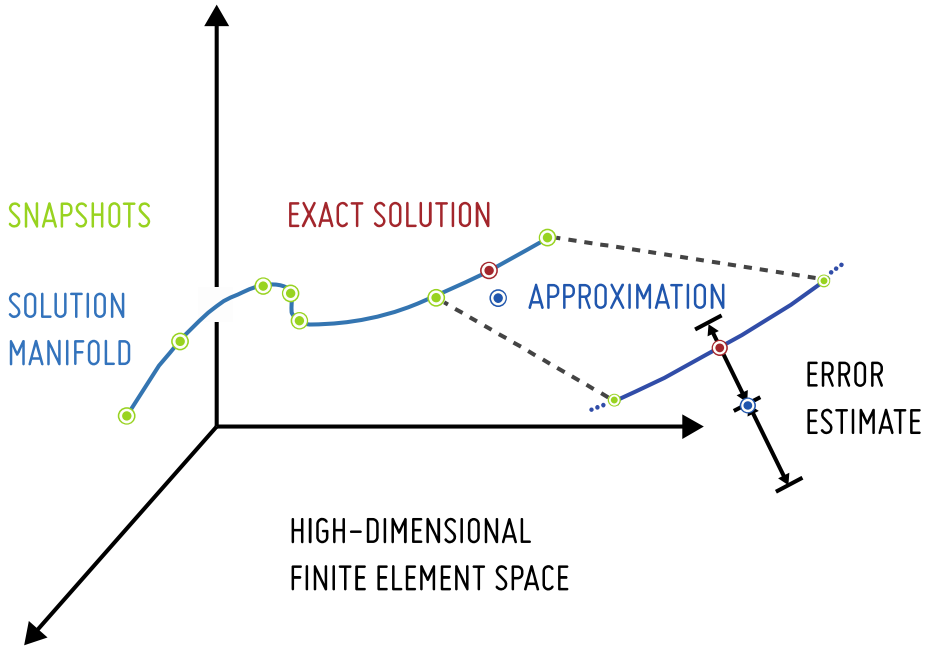
3. Perform the POD with pre-defined tolerance

$$\frac{\sum_{i=1}^{r} \lambda_i^2}{\sum_{i=1}^{N} \lambda_i^2} \leq \varepsilon$$

   - Input: training snapshots
   - Output: basis functions

4. Perform the projection (machine learning)
   - Input: training parameters (e.g., material properties)
   - Output: reduced coefficients

5. Back-Projection: $u_{rb}(\theta) = \sum_{i=1}^{r} \beta_{rb}^{(i)}(\theta)\psi_i \in V_{rb}$



Degen et al. (2020), Degen et al. (2023), Hesthaven et al. (2018)

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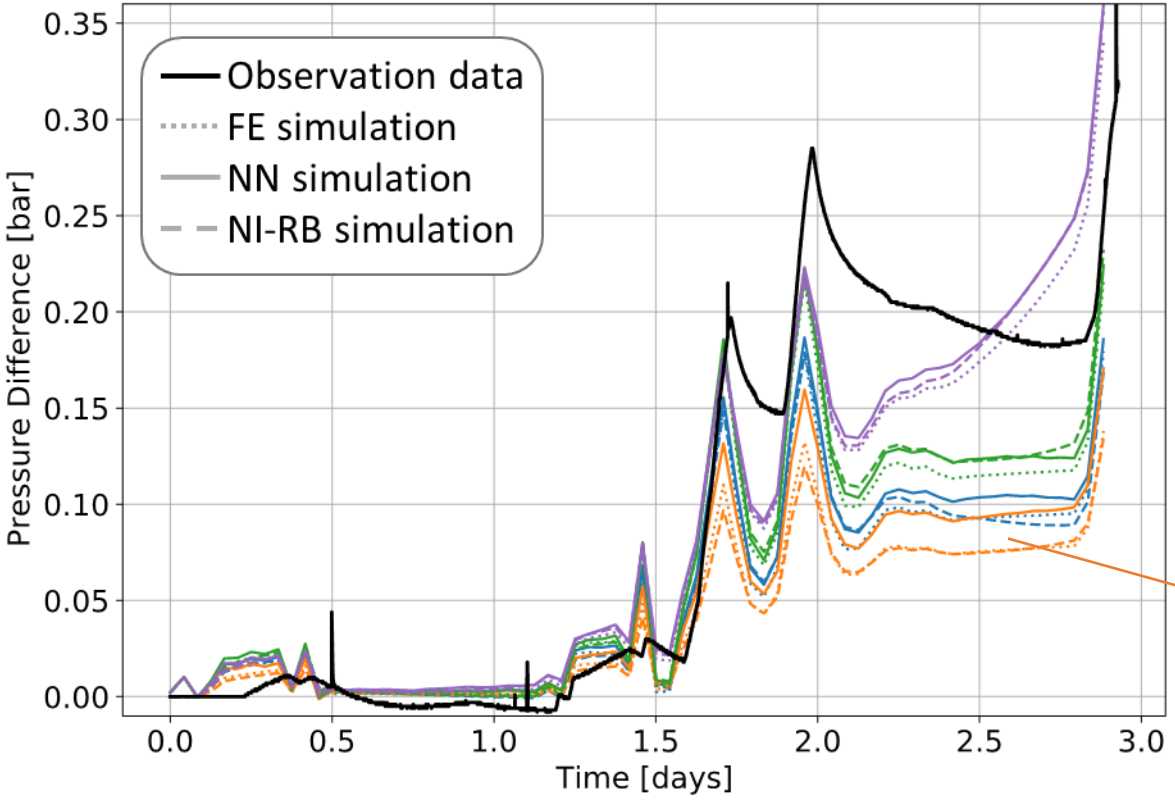
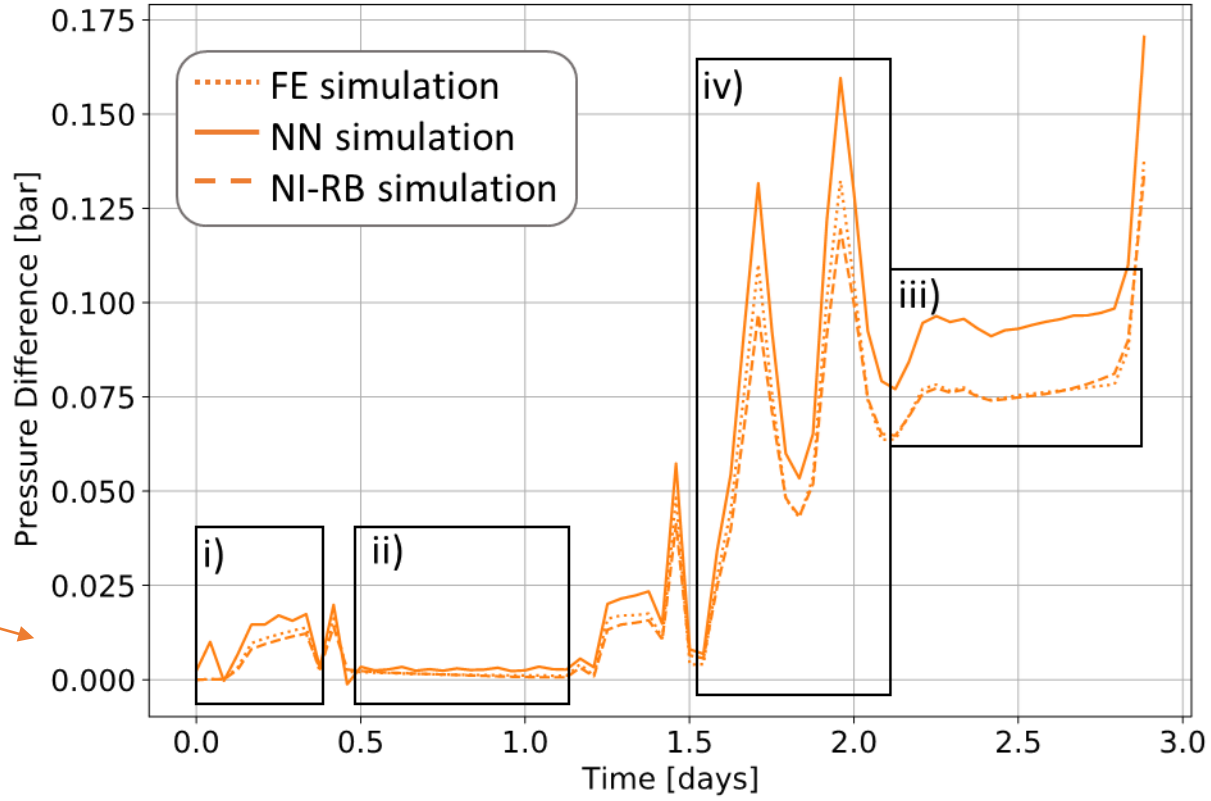
RELIABLE AND INTERPRETABLE MODELS





Degen et al. (2024)

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BENCHMARK – UTAH FORGE PROJECT

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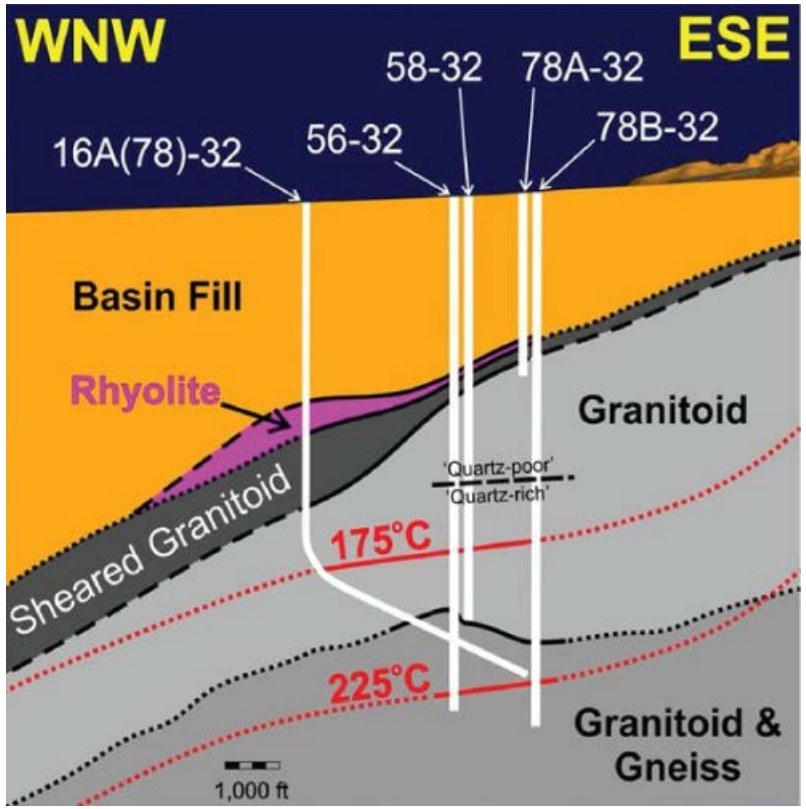
UTAH FORGE PROJECT

- Forge serves as a test platform to validate the commercial replicability of geothermal energy extraction
- Main objective is to identify and quantify the key processes controlling Enhanced Geothermal System (EGS) performance
- The site conditions:
  - Temperature in the range ~175-225 °C at ~2-3 km depth
  - Low permeability granite ~10^{-16} m²
  - Good ‘hot dry rock’ EGS target



Data Access: https://utahforge.com/

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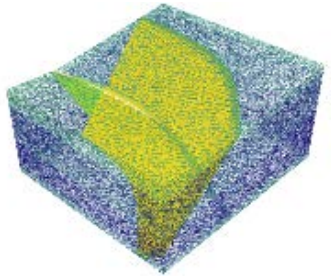
DATA SET GENERATION & WORKFLOW

Mesh geometry, faults&well structure → Coupled thermo-hydraulic model [FOM] → Surrogate model

Tools:

GOLEM



PyMeshIt

Developed by GECOS-LAB

Example of a surrogate model set-up:

Metric	*Forge*
Samples Training	100
Samples Validation	20
Sampling Method Training	Latin-Hypercube
Sampling Method Validation	Random
Scaling Parameters	Z-score
Scaling Data Set	Min-Max

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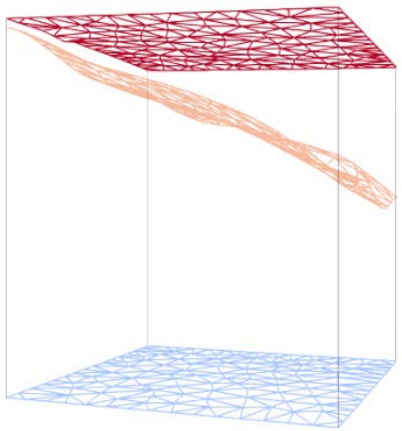
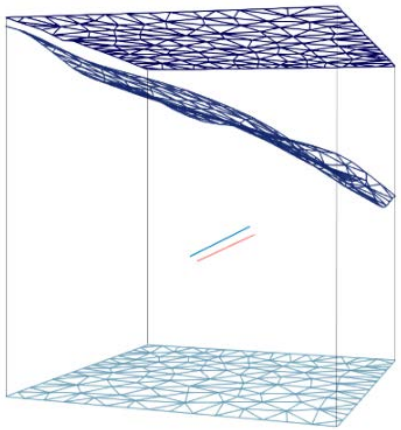
GEOTHERMAL MODELING STAGES

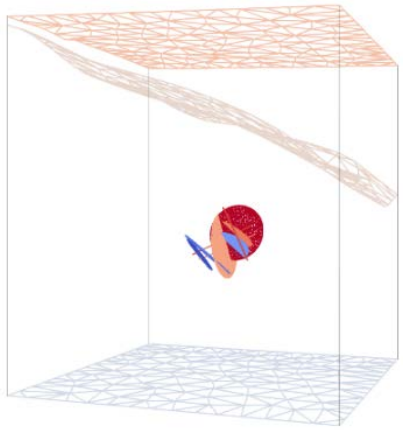
Geological knowledge / Data availability



Scenario 1:
Sedimentary rocks and granite units



Scenario 2:
Injection (16A) and production (16B) wells



Scenario 3:
Injection, production wells and 7 fractures

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SCENARIO 1 – GEOLOGICAL UNITS

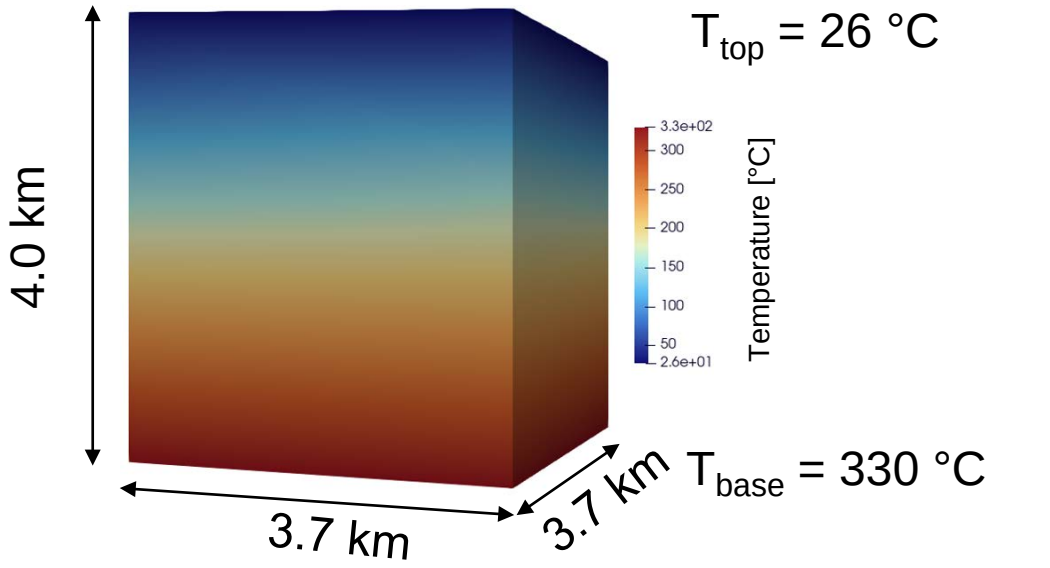
- Steady state solutions
  - For temperature

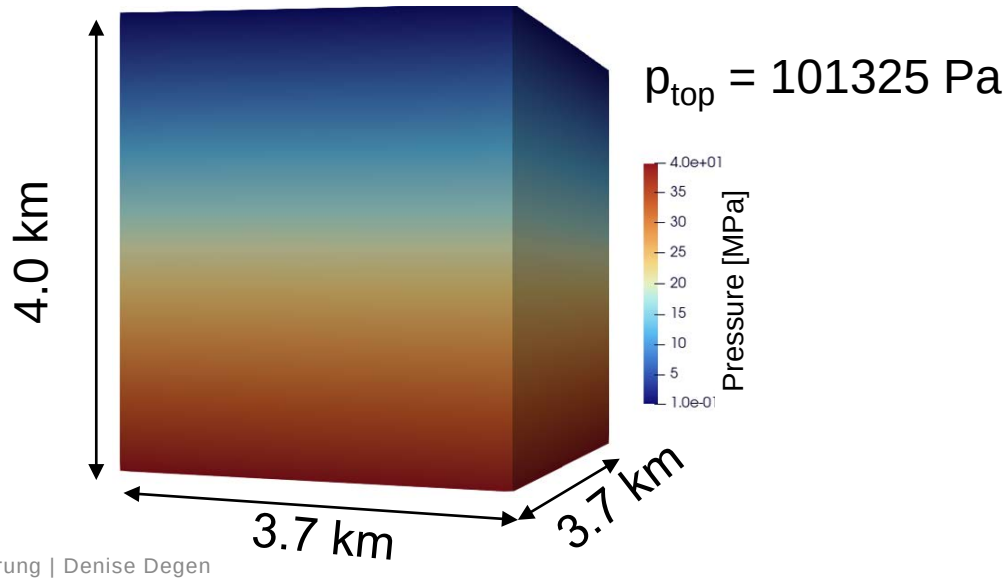
$$\nabla \cdot (\lambda_b \nabla T) = 0$$

  - And pressure

$$q_D = -\frac{k}{\mu_f}\left(\nabla p_f - \rho_f g\right)$$

- Boundary conditions:
  - Dirichlet boundary conditions for the temperature at the top and base of the model
  - Dirichlet boundary condition for the pressure at the top of the model





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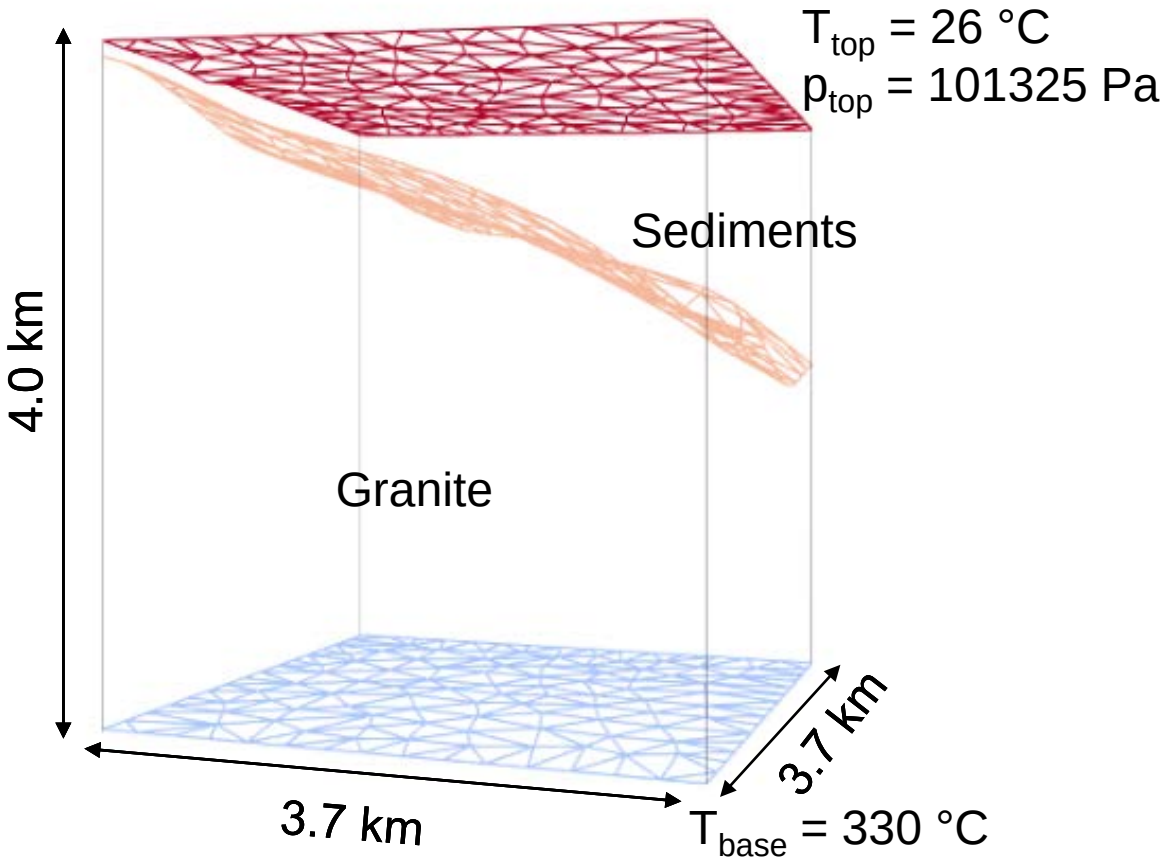
SCENARIO 1 – PROPERTIES

Units	ϕ [%]	k [m²]	Flow rate [kg/s]
Sediments	12	10^{-14}	n/a
Granite	0.1-1.0	10^{-14}-10^{-16}	n/a



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SCENARIO 2 – WELLS

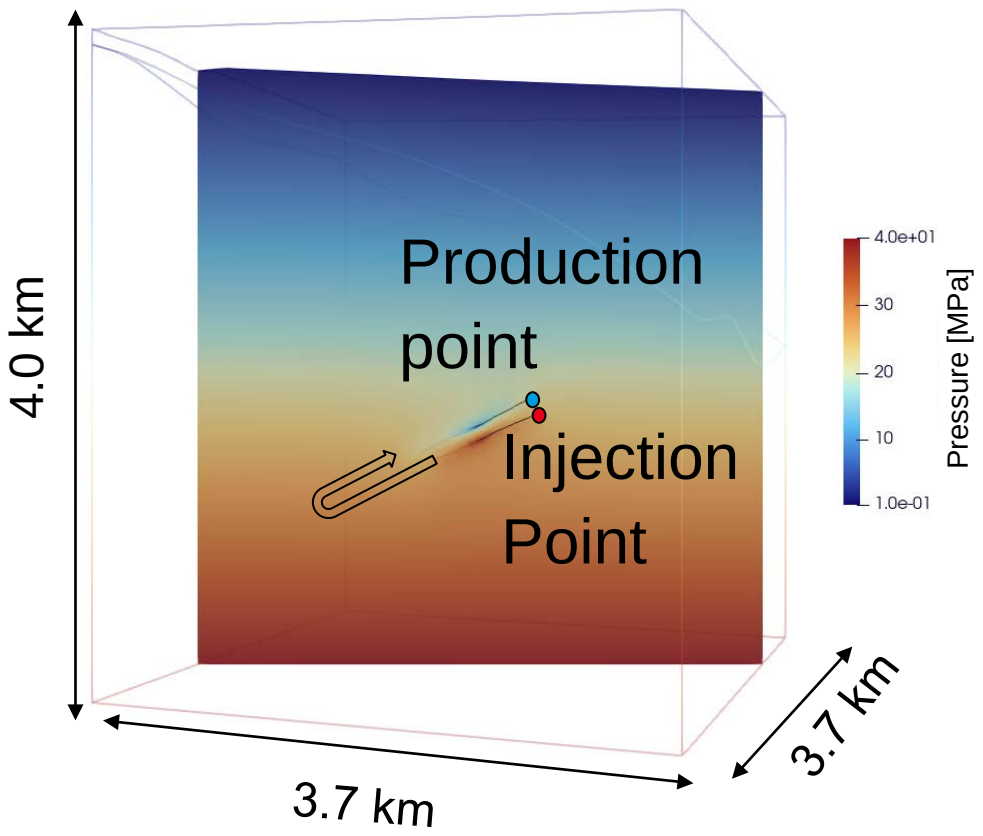
- Coupled hydrothermal simulations

$$(\rho c)_b \frac{\partial T}{\partial t} + \nabla \cdot \left((\rho c)_f q_D T - \lambda_b \nabla T \right) = 0,$$

$$\frac{1}{M_b} \frac{\partial p_f}{\partial t} + \nabla \cdot q_D = 0,$$

$$with\ q_D = -\frac{k}{\mu_f} \left(\nabla p_f - \rho_f g \right).$$

- Boundary conditions:
  - Dirichlet boundary conditions for the temperature at the top and base of the model
  - Dirichlet boundary condition for the pressure at the top of the model



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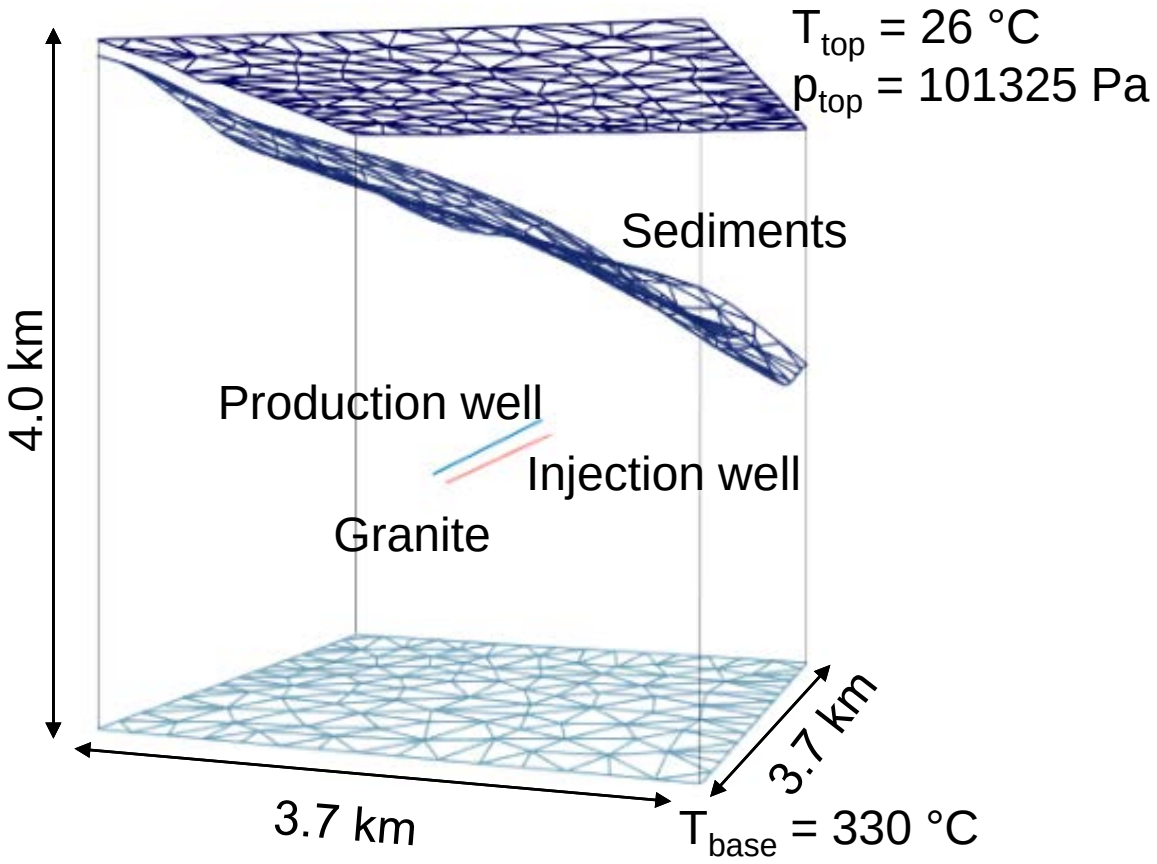
SCENARIO 2 – PROPERTIES

Units	ϕ [%]	k [m²]	Flow rate [kg/s]
Sediments	12	10^{-14}	n/a
Granite	0.1-1.0	10^{-14}-10^{-16}	n/a
Wells	n/a	n/a	12.0-40.0



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SCENARIO 3 – FRACTURES

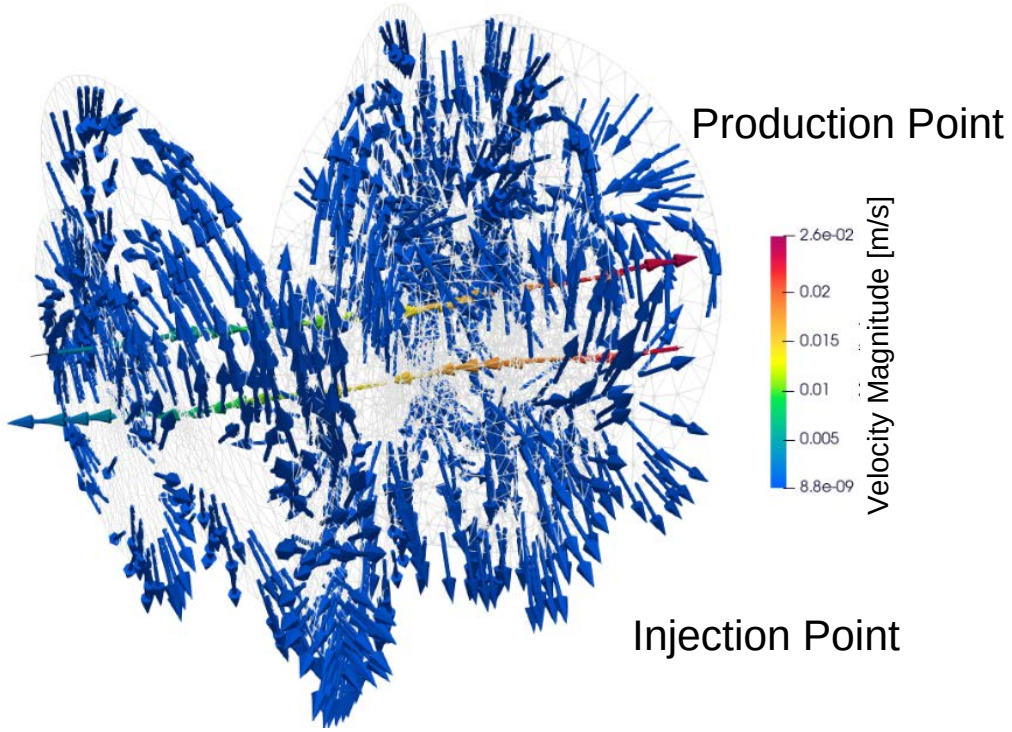
- Coupled hydrothermal simulations

$$(\rho c)_b \frac{\partial T}{\partial t} + \nabla \cdot \left((\rho c)_f q_D T - \lambda_b \nabla T \right) = 0,$$

$$\frac{1}{M_b} \frac{\partial p_f}{\partial t} + \nabla \cdot q_D = 0,$$

$$with\ q_D = -\frac{k}{\mu_f} \left(\nabla p_f - \rho_f g \right).$$

- Fracture integration to the model:
  - 2D planes representing major fractures, produced during hydraulic stimulation → EGS reservoir
  - Injection and productions rates are balanced
  - Possible extension → THM transient model, induced seismicity



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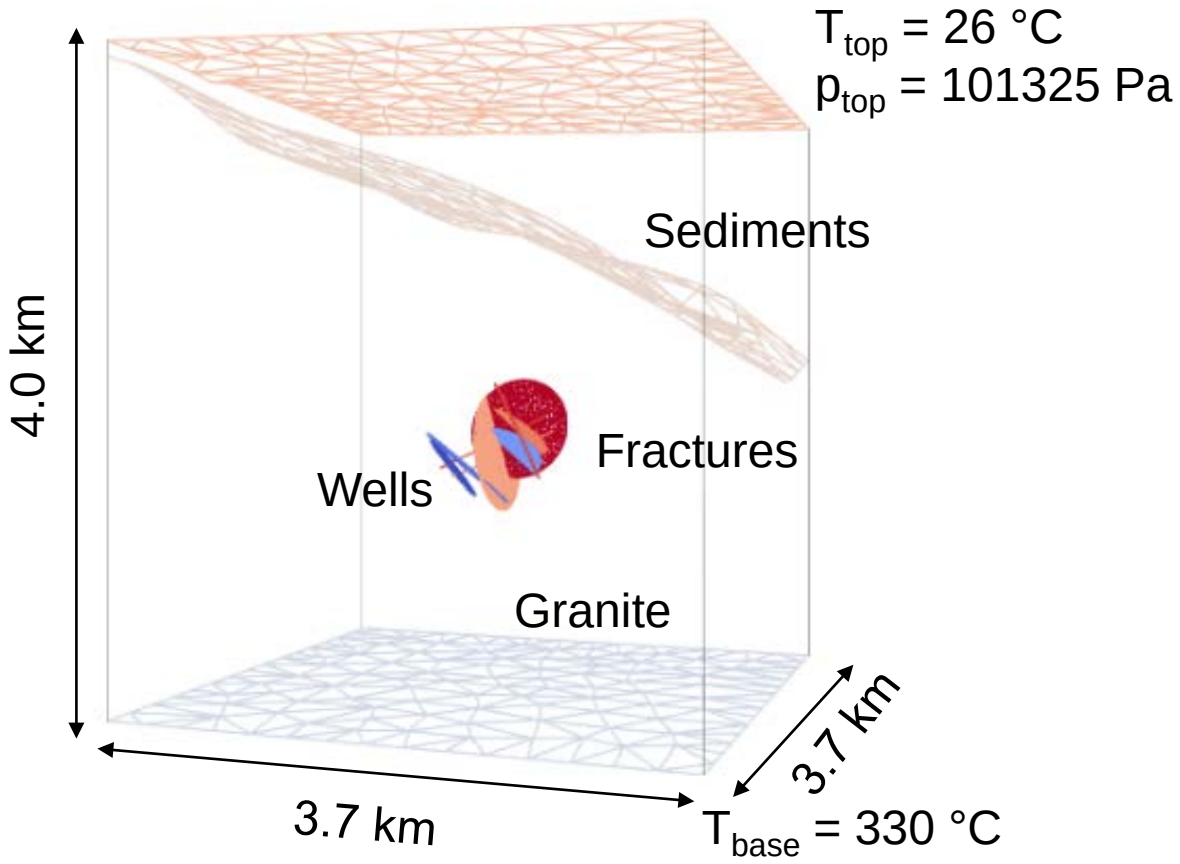
SCENARIO 3 – PROPERTIES

Units	ϕ [%]	k [m²]	Flow rate [kg/s]
Sediments	12	10^{-14}	n/a
Granite	0.1-1.0	10^{-14}-10^{-16}	n/a
Wells	100	n/a	12.0-40.0
Fractures	100	10^{-13}-10^{-14}	n/a

- Fracture hydraulic properties such as aperture and permeability can be estimated as:

Aperture [mm]	$0.607\ r_{equivalent}^{2}$
k[m²]	$1.66\ aperature^{1.5}$



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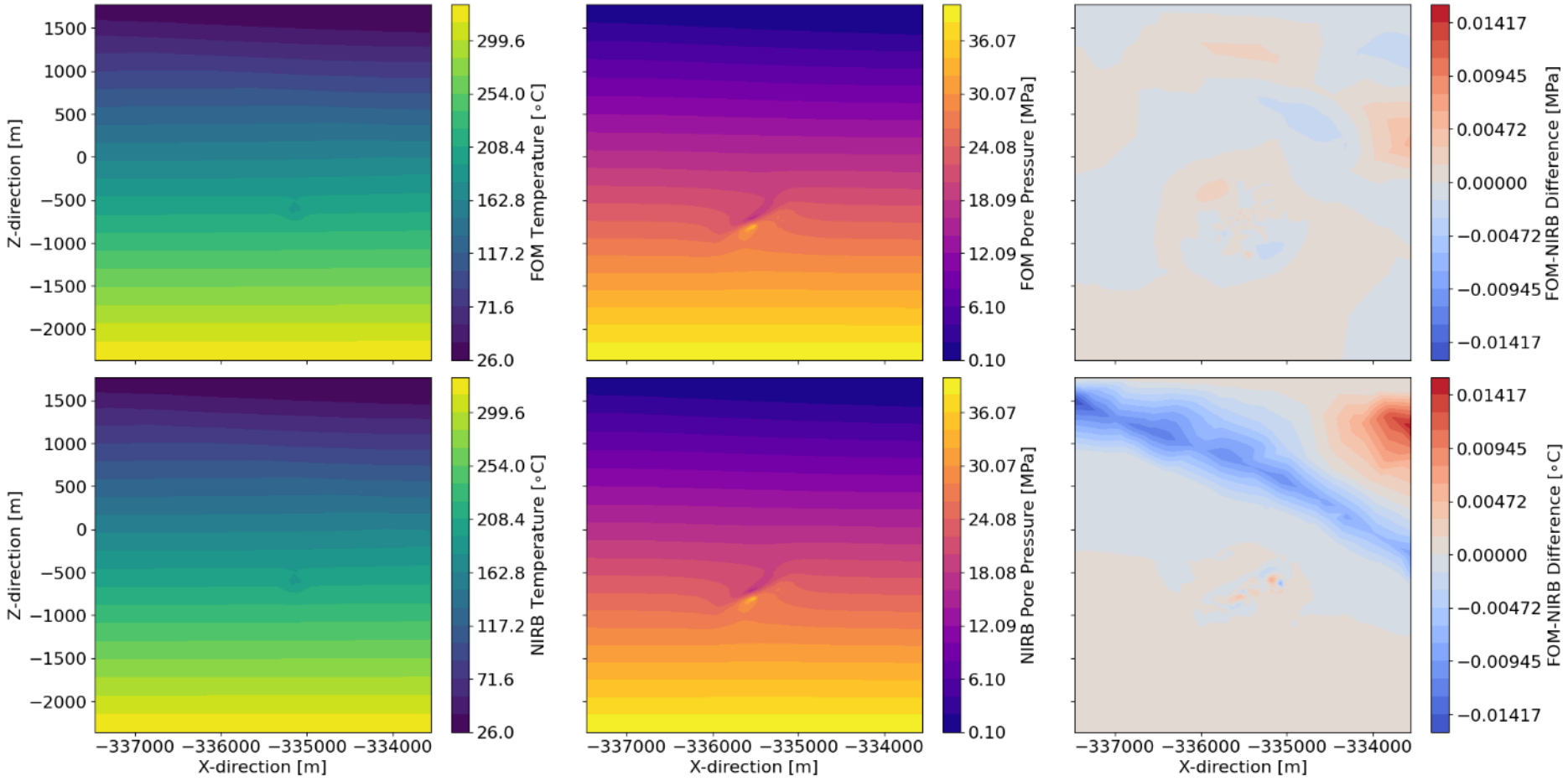
ERROR - METRICES

- Global Errors
  - R^2
  - RMSE
- Local error behavior
  - Representative 2D cross section
  - Representative 1D line



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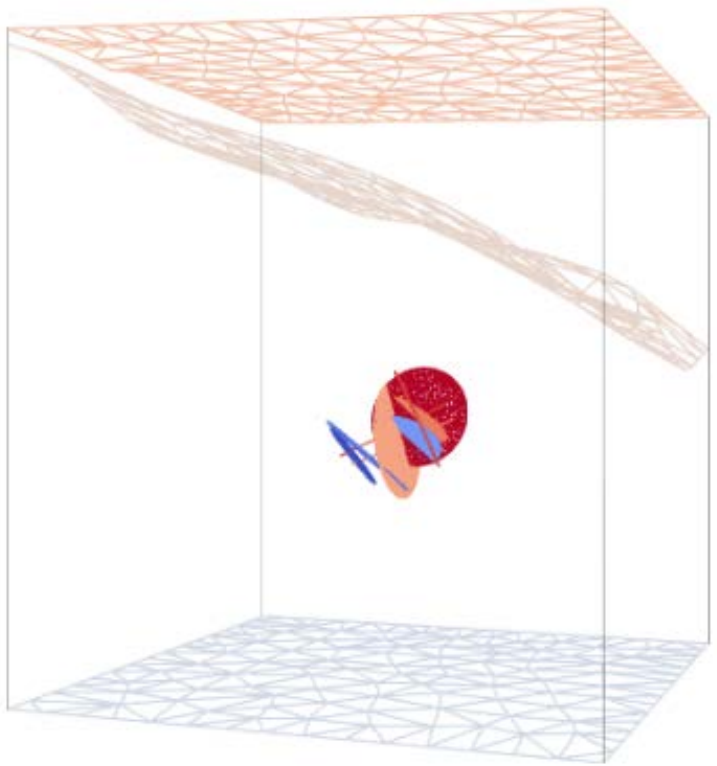
CONCLUSIONS

- Benchmark for parameter estimation studies
- Including different levels of physical and geological complexity
- Designed for methods with a focus on data sparse applications
- Focus on physical preservation



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